

Introduction to CVS, Overview of thoracic cavity and Mediastinum

Outline

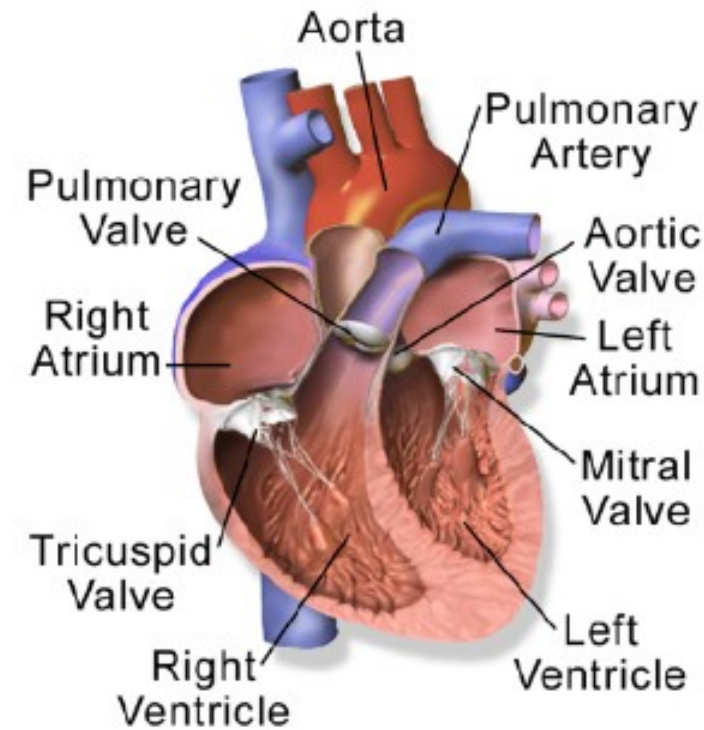
- Introduction to Cardiovascular System
- Overview of thoracic cavity
- Mediastinum
 - Superior and Anterior Mediastinum

What is the Cardiovascular System

- The **cardiovascular system**, also called the **circulatory system**, is the organ system that transports materials to and from all the cells of the body.
- The materials carried by the cardiovascular system include **oxygen** from the **lungs**, **nutrients** from the **digestive system**, **hormones** from glands of the **endocrine system**, and **waste** materials from **cells throughout the body**.
- Transport of these and many other materials is necessary to maintain homeostasis of the body.
- The main components of the cardiovascular system are the **heart**, **blood vessels**, and **blood**.

The Heart

- The heart is a muscular organ in the thoracic cavity.
- It consists mainly of cardiac muscle tissue and pumps blood through blood vessels by repeated, rhythmic contractions.
- The heart has four chambers: a **right atrium** and **right ventricle** and a **left atrium** and **left ventricle**.
- On each side of the heart, blood is pumped from the **atrium** to the **ventricle** and from the ventricle **out of the heart / to the blood vessels**.



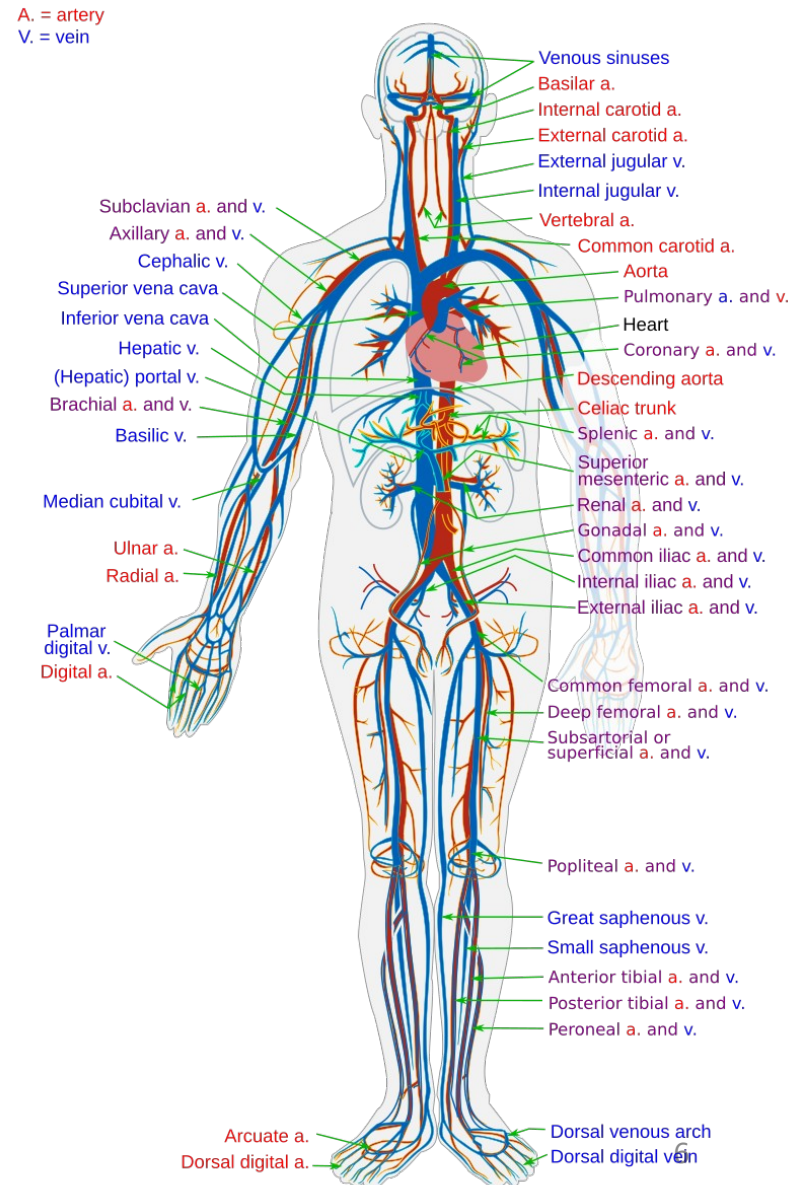
Anatomy of the Heart

The Heart

- The heart also contains **valves** that allow **blood to flow** only in the proper **direction** through the heart.
- Unlike skeletal muscle, cardiac muscle routinely contracts **without stimulation** by the nervous system.
- Specialized cardiac muscle cells send out electrical impulses that stimulate the contractions.
- **Location:**
 - Superior surface of diaphragm
 - Left of the midline
 - Anterior to the vertebral column, posterior to the sternum

Blood Vessels

- Like a network of roads, the blood vessels have the job of allowing the transport of materials from one place to another.
- There are three major types of blood vessels: **arteries**, **veins**, and **capillaries**.
- **Arteries** are blood vessels that carry blood away from the heart (except for the arteries that actually supply blood to the heart muscle).

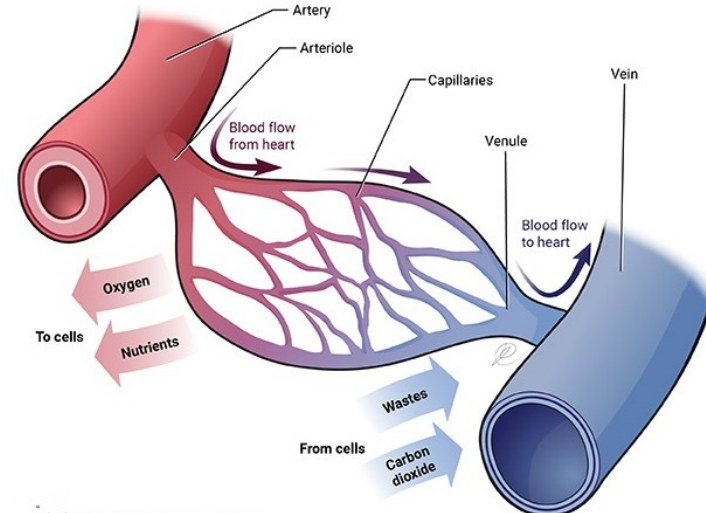


Blood Vessels...

- Most arteries **carry oxygen-rich blood**, and one of their main functions is **distributing oxygen (except _____ arteries)** to tissues throughout the body.
- The smallest arteries are called **arterioles**.

Blood Vessels...

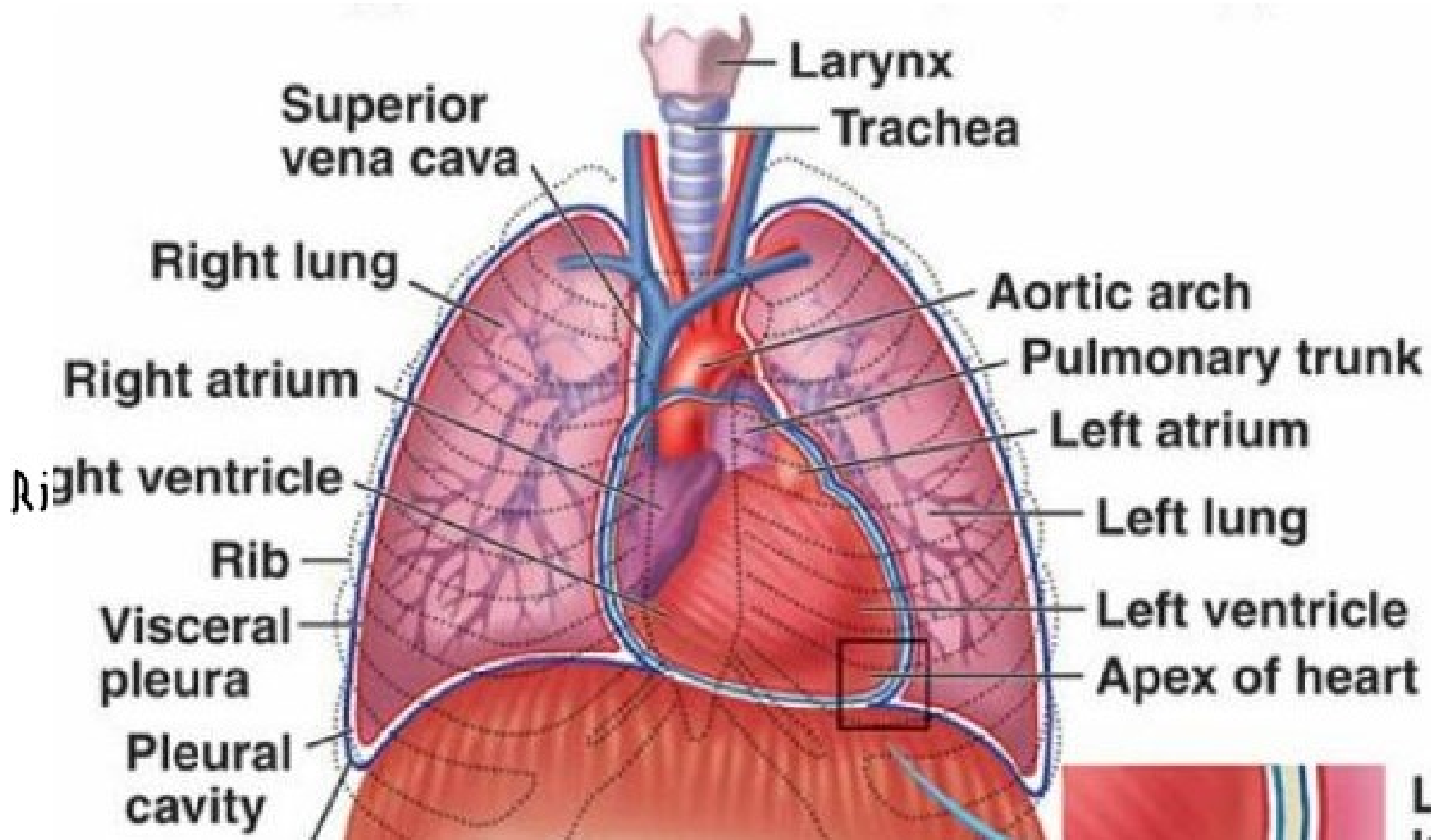
- Veins are blood vessels that **carry blood toward the heart.**
- Most veins carry **deoxygenated blood.**
- The smallest veins are called **venules.**
- **Capillaries** are the smallest blood vessels.
- They connect arterioles and venules.
- As they pass through tissues, they exchange **substances** including oxygen with cells.



Thoracic Cavity

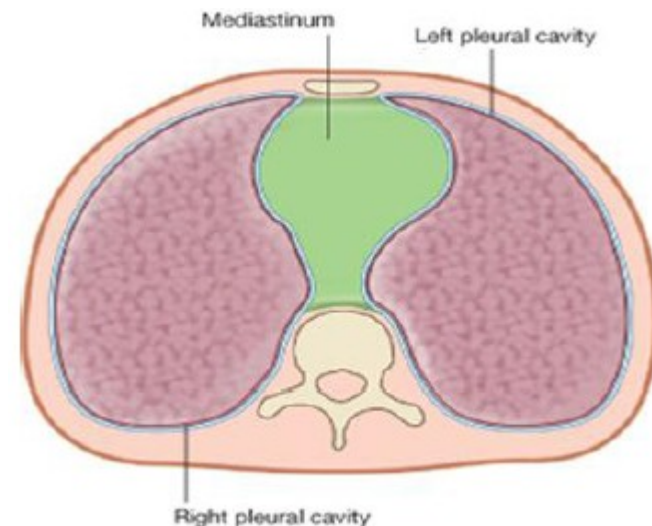
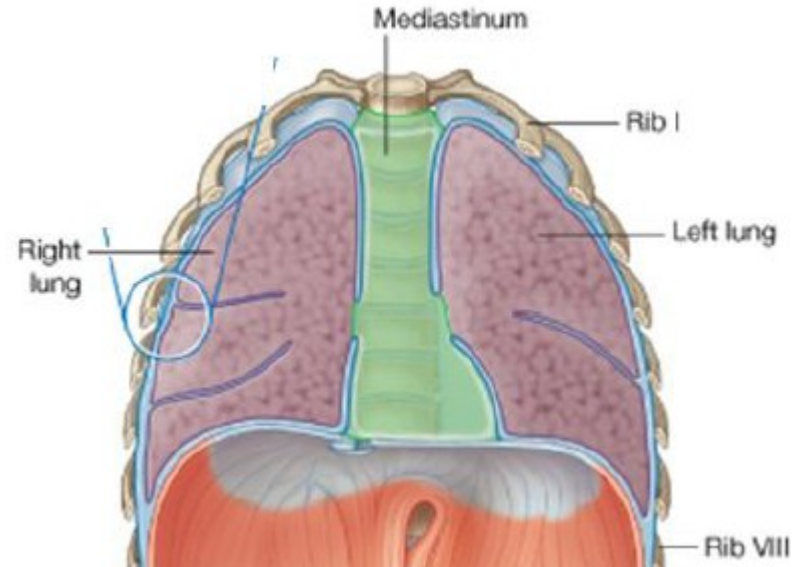
- The thoracic cavity is found deep to the thoracic wall, superior to the **diaphragm**, and inferior to the root of the neck (**thoracic aperture**).
- The thoracic cavity contains organs and tissues that function in the
 - respiratory (lungs, pleura, bronchi, trachea,)
 - cardiovascular (heart, pericardium, great vessels, lymphatics),
 - nervous (vagus nerve, sympathetic chain, phrenic nerve, recurrent laryngeal nerve),
 - immune (thymus) and
 - digestive systems (esophagus).

Thoracic Cavity...



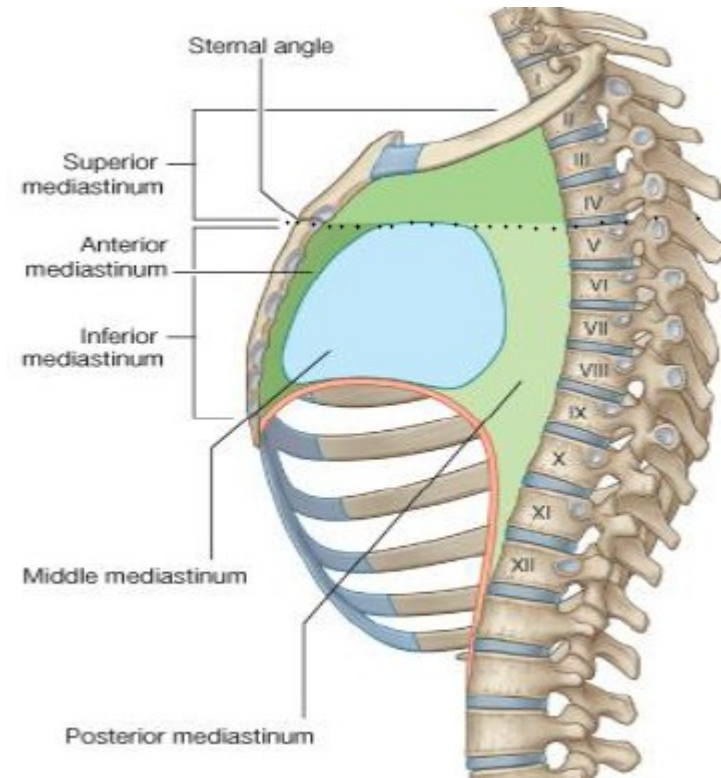
Thoracic Cavity...

- The thoracic cavity can typically be divided into well-established compartments.
 - **Pleural cavities and**
 - **Mediastinum**
- There are **two pleural** cavities which respectively contain the left and right **lungs** and **pleura**.
- Median region b/n the **two pleural cavities**
- Extends from **superior thoracic aperture** to **diaphragm** and from **sternum** to **bodies of thoracic vertebrae**



Mediastinum

- Through the **angle of the sternum** a horizontal plane crossing the T4-T5 vertebral junction (IVD b/n T4 & T5) and divides the mediastinum in to
 - ✓ **Superior mediastinum**
 - ✓ **Inferior mediastinum**
- Inferior mediastinum** is further subdivided by the **pericardium** into
 - ✓ **Anterior mediastinum**
 - ✓ **Middle mediastinum and**
 - ✓ **Posterior mediastinum**



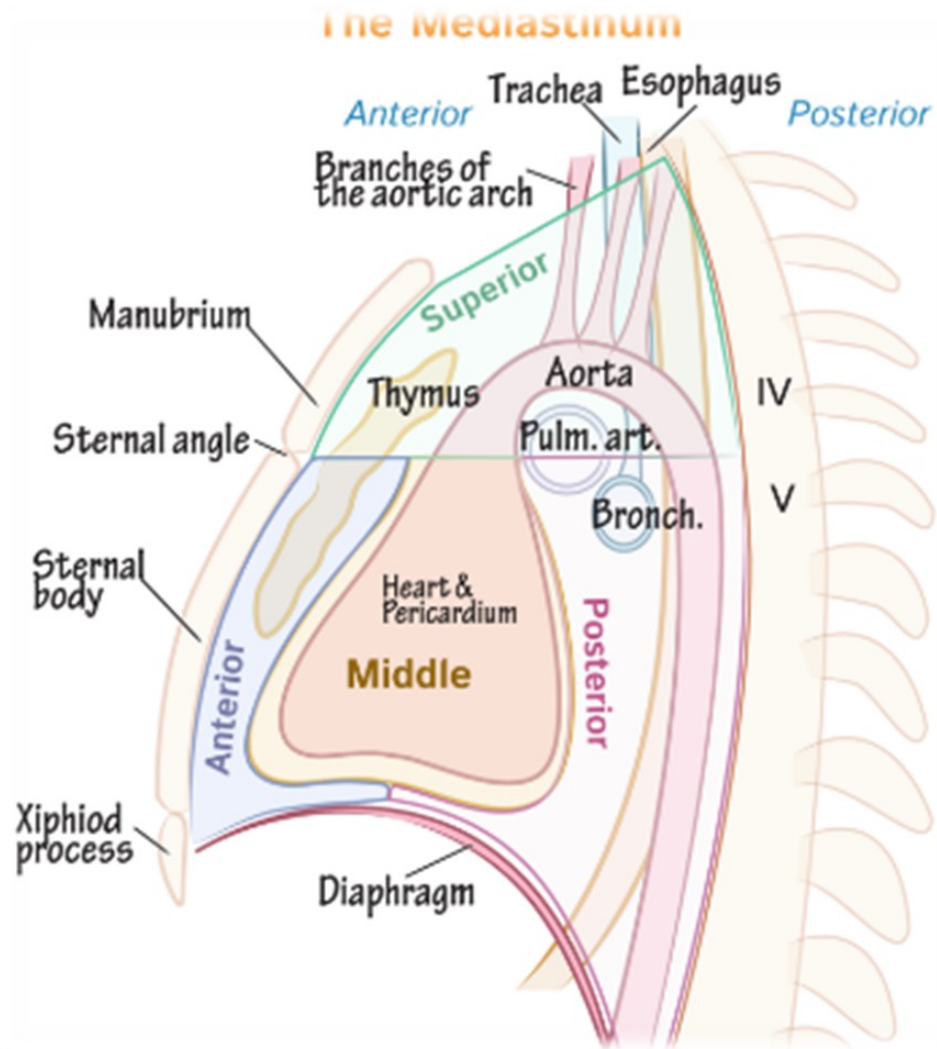
Thoracic Cavity...

- **Anterior mediastinum** is anterior to the pericardial sac
- **Middle mediastinum** contains the heart and pericardium
- **Posterior mediastinum** is posterior to the pericardial sac.

Thoracic Cavity

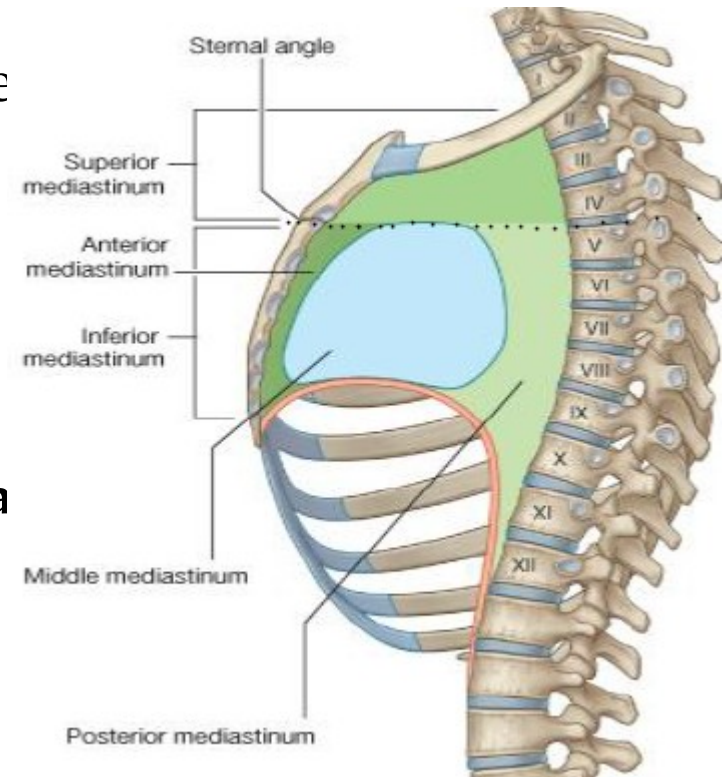
- Gross contents of each mediastinal compartment are:
 - **Superior mediastinum** - great vessels, trachea, esophagus, vagus nerve, phrenic nerve, sympathetic nerves, thoracic lymphatic duct, thymus
 - **Anterior mediastinum** - CT, thymus, and lymph nodes
 - **Middle mediastinum** - heart, pericardium, roots of great vessels and phrenic nerve
 - **Posterior mediastinum** - descending aorta, azygos veins, esophagus, vagus nerve, sympathetic nerves, bronchi, and thoracic lymphatic duct

Thoracic Cavity...



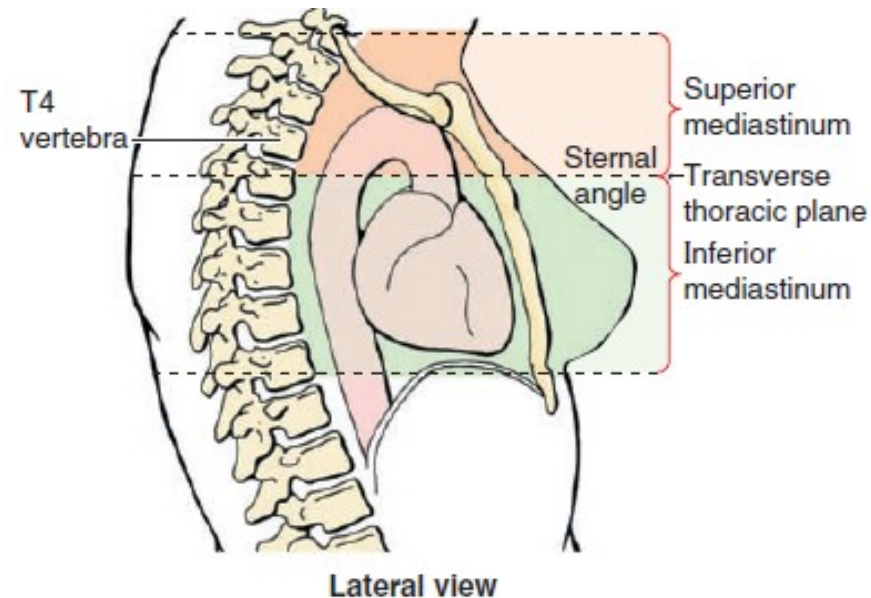
Anterior mediastinum

- the smallest subdivision of the mediastinum
- lies between the body of the sternum the pericardium **posteriorly**.
- It is continuous from sternal angle the diaphragm **inferiorly**.
- It consists of
 - loose connective tissue (**sternopericardial ligaments**)
 - fat, lymphatic vessels, and a few lymph nodes
 - branches of the internal thoracic vessels
 - inferior part of the thymus



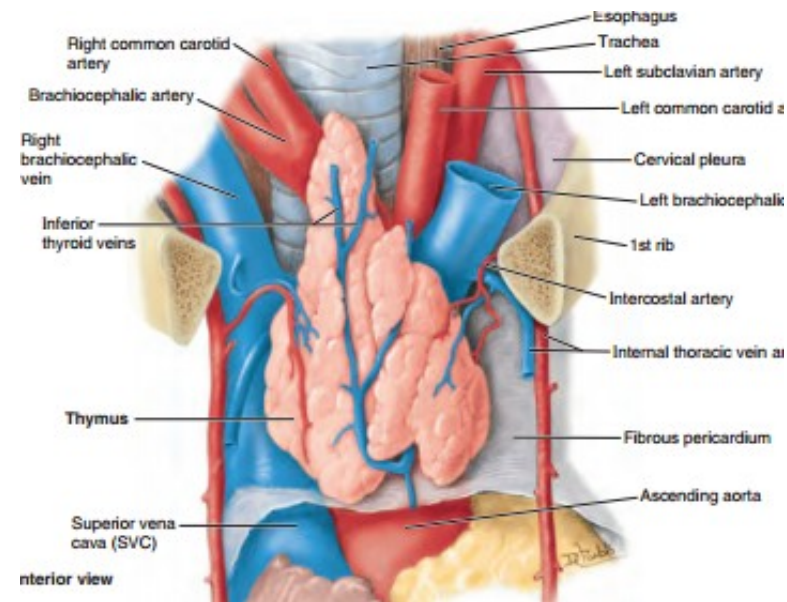
Superior mediastinum

- The **superior mediastinum** is superior to the transverse thoracic plane
- Bounded in front by the **manubrium** and behind by the first 4 thoracic vertebrae
- Continuous superiorly with the neck and inferiorly with the inferior mediastinum



Superior mediastinum

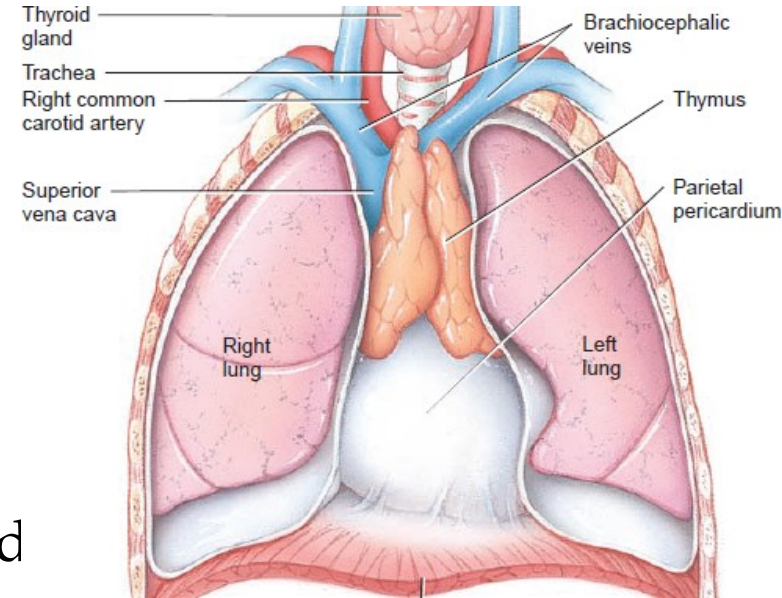
- **Contents** superior mediastinum from anterior to posterior are
 - **Thymus**
 - **Great veins** - brachiocephalic veins & SVC
 - **Great arteries** - arch of aorta, the brachiocephalic trunk, left common carotid artery, and left subclavian artery
 - **Nerves** - vagus and phrenic nerves & cardiac plexus
 - Thoracic part of **trachea**
 - **Esophagus**
 - **thoracic duct** and **lymphatic trunks**.



Glandular plane

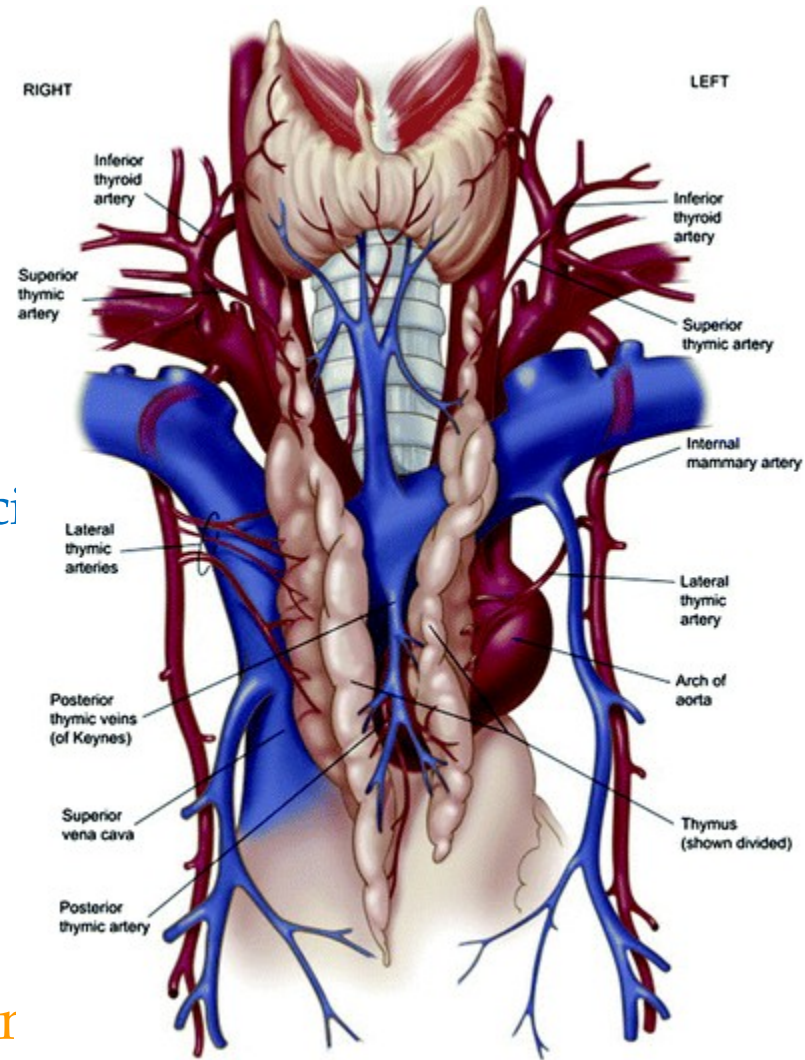
Thymus

- The most **anterior** structure immediately posterior to the **manubrium**
- Prominent in **children**
- extends from lower part of the thyroid gland to the **4th costal cartilage**
- It consists of **two lobes** and is **mainly fat in the adult** with small islets of active thymic cells scattered throughout.



Thymus...

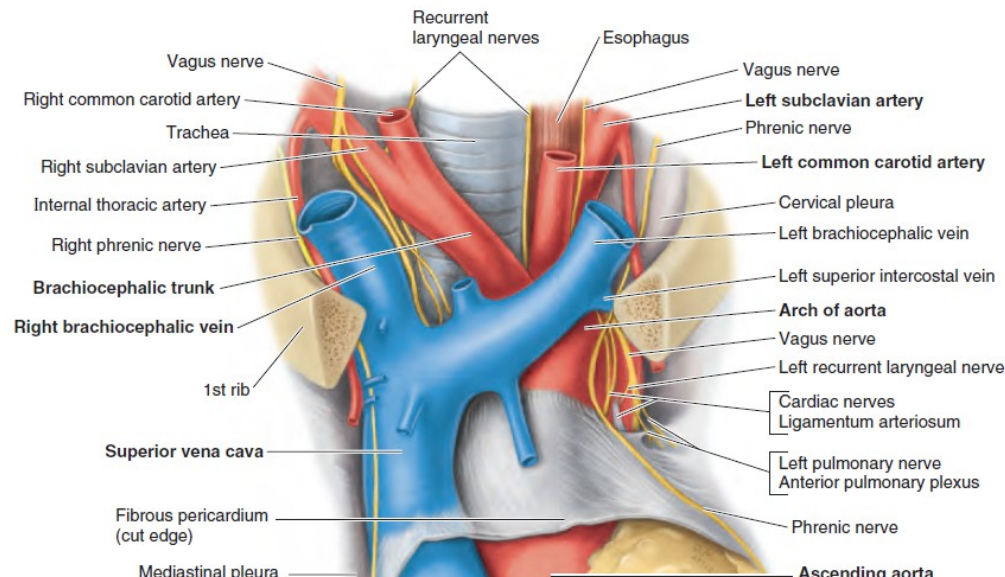
- Arterial supply –
 - inferior thyroid, internal thoracic, anterior intercostal
- Venous drainage –
 - to left brachiocephalic, internal thoracic & anterior intercostal
- Lymphatics –
 - parasternal, brachiocephalic & trachiobronchial
- Nerves – sympathetic nervous system and Vagus nerve



Venous plane

Left BCV and Right BCV

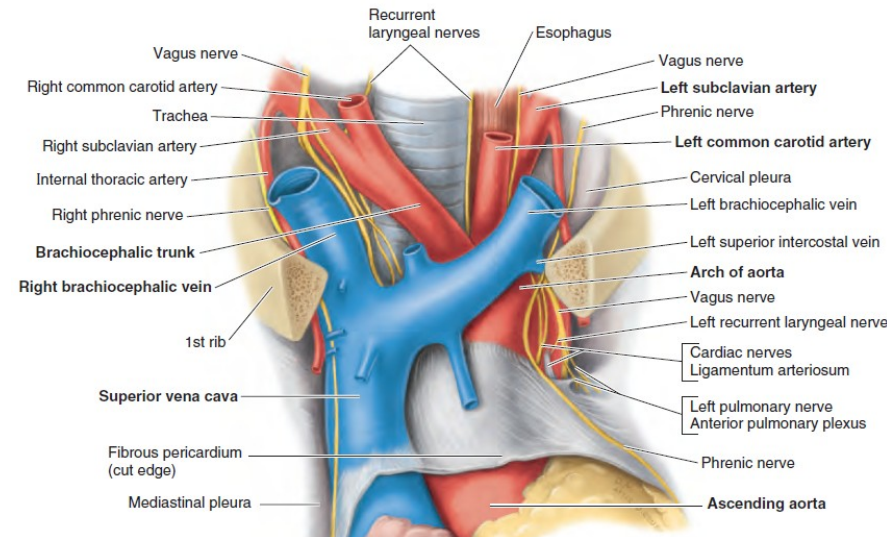
- ✓ Formed posterior to the sternoclavicular joints by the union of the internal jugular and subclavian veins
- ✓ At the level of the **inferior border** of the **1st right costal cartilage**, the brachiocephalic veins unite to form the **SVC**
- ✓ The brachiocephalic veins shunt blood from the **head, neck, and upper limbs** to the **SVC then to** right atrium.



Venous plane....

Superior vena cava (SVC)

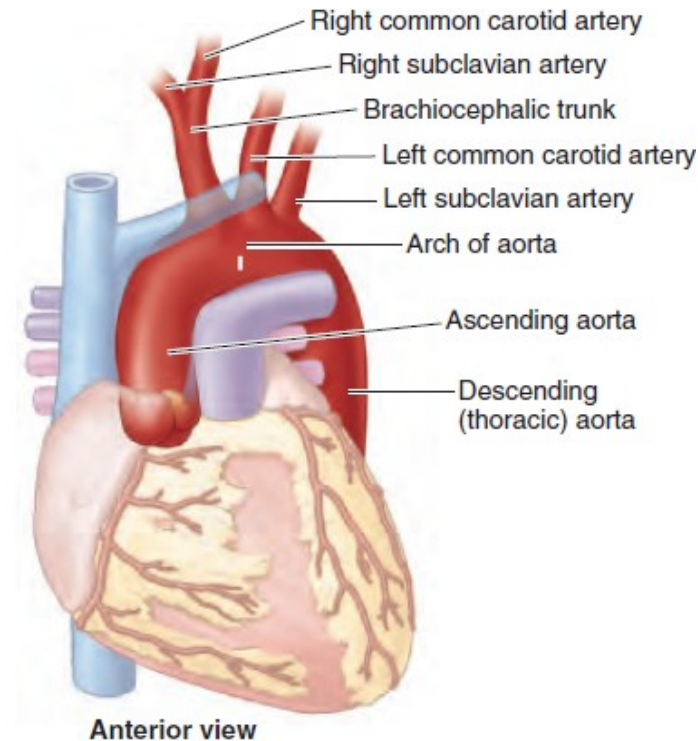
- The SVC lies in the right side of the superior mediastinum, anterolateral to the trachea and posterolateral to the ascending aorta.
- Returns blood from all structures superior to the diaphragm, except the lungs and heart.
- At the level of the 2nd costal cartilage, the SVC enters the middle mediastinum
- ends at the level of the 3rd costal cartilage, where it enters the right atrium



Arterial plane

Arch of aorta

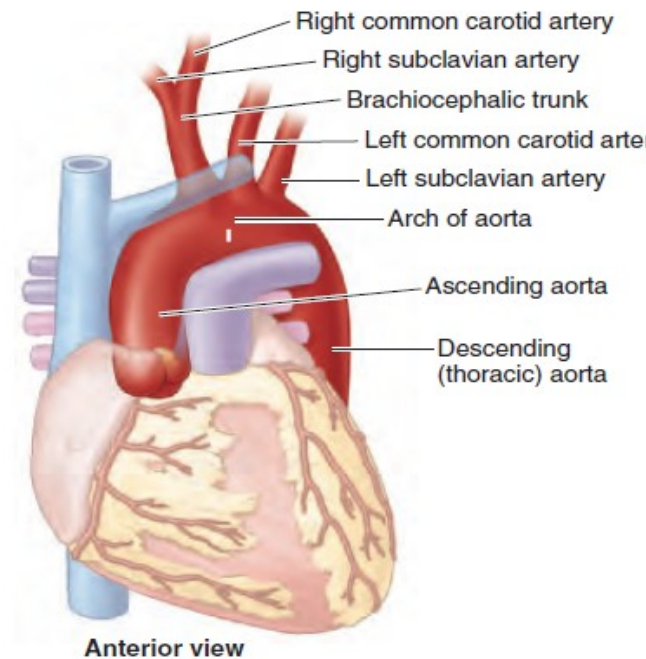
- the curved continuation of the ascending aorta
- The arch begins posterior to the 2nd **right** sternocostal (SC) joint at the level of the sternal angle
- ends by becoming the **descending aorta** posterior to the **2nd left** sternocostal joint
- The arch descends posterior to the left root of the lung beside the T4 vertebra



Branches of aortic arch

❖ Brachiocephalic trunk

- Largest branch of arch of aorta
- it is **anterior to the trachea** and posterior to the **left brachiocephalic vein**
- It ascends superolaterally to reach the right side of the trachea and it divides into the **right common carotid** and **right subclavian artery**



❖ Left Common Carotid

- the second branch of the arch of the aorta
- is at first anterior to the trachea and then to its left.

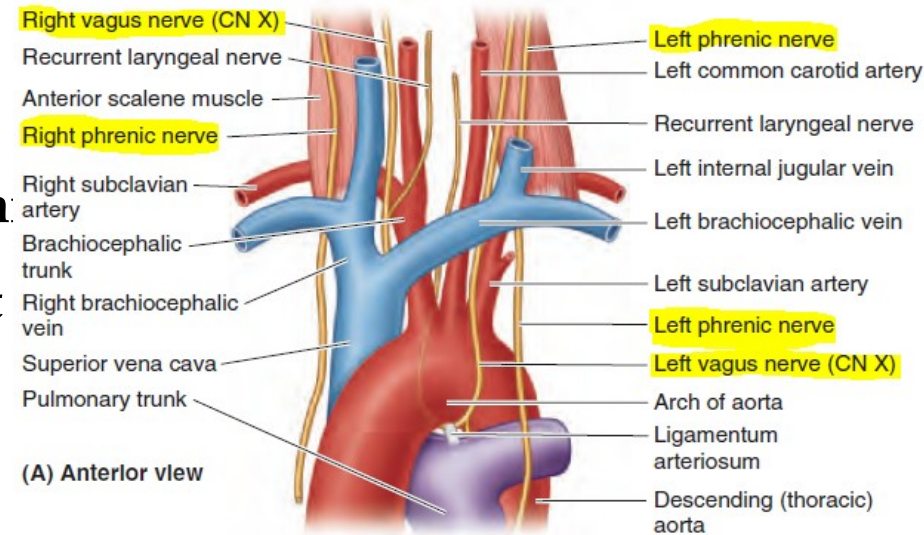
❖ Left Subclavian A.

- the third branch of the arch of the aorta
- arises from the posterior part of the arch
- It ascends lateral to the trachea and left **common carotid artery** through the superior mediastinum

Nerves in the superior mediastinum

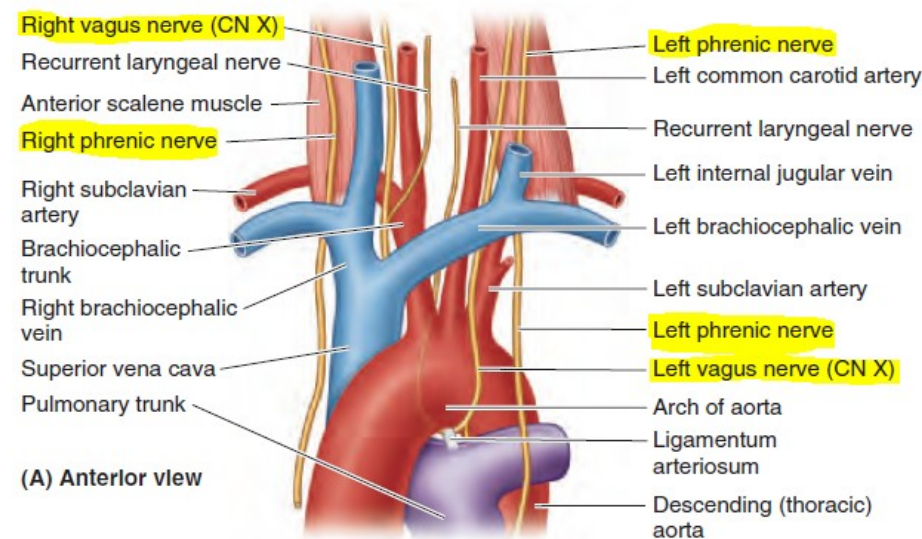
Vagus nerve

- **Right vagus nerve** enters the thorax **anterior** to the **right subclavia** artery, where it gives rise to the **right recurrent laryngeal nerve**
- The right recurrent laryngeal nerve **hooks** around the **right subclavian artery** and ascends b/n the **trachea** and **esophagus** to supply the larynx.
- Contributes fibers to the **esophageal plexus** also gives rise to nerves that contribute to the **cardiac plexus**.



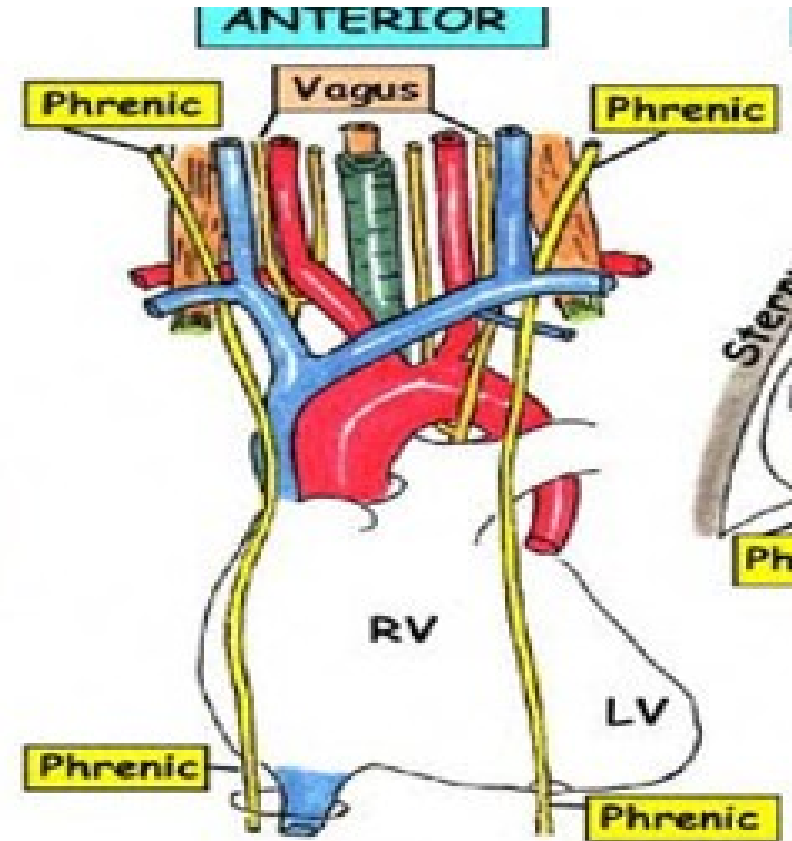
Left vagus nerve (LVN)

- It descends in the neck **posterior** to the **left common carotid artery**
- It enters the mediastinum between the **left common carotid artery** and **left subclavian artery**.
- As the LVN curves medially at the inferior border of the **arch** of the **aorta**, it gives off the **left recurrent laryngeal nerve**.



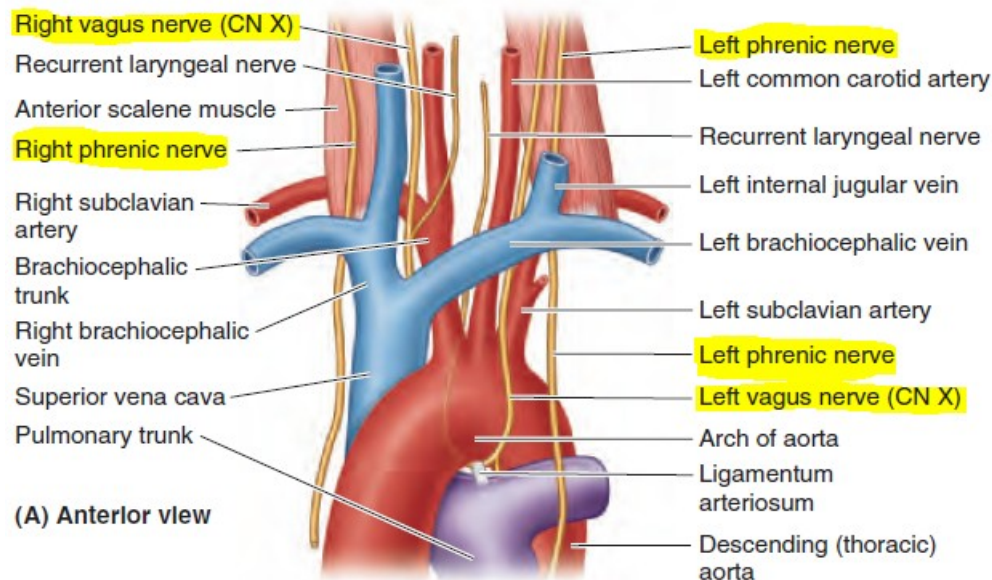
Left vagus nerve (LVN)

- The left recurrent laryngeal nerve passes inferior to the arch of the aorta and ascends between the trachea and the esophagus and supply intrinsic muscle of the larynx
- The LVN passes **posterior to the root of the left lung**, where it breaks up into many branches
- ❖ Rt and Lt vagus nerve passes **posterior to the root** of the lung unlike phrenic nerve.



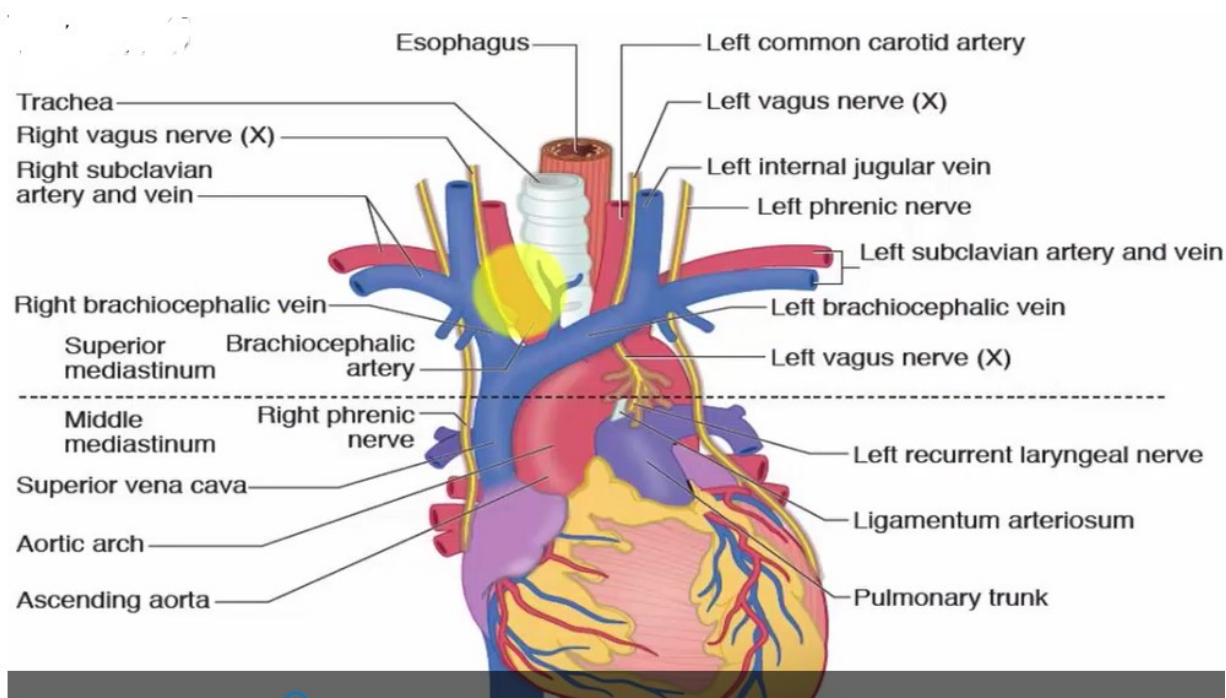
Phrenic nerve

- Is from ventral rami of C3, C4 & C5
- Each phrenic nerve enters the superior mediastinum between the **subclavian artery** and the origin of the **brachiocephalic vein**
- They pass **anterior** to the **roots** of the lungs



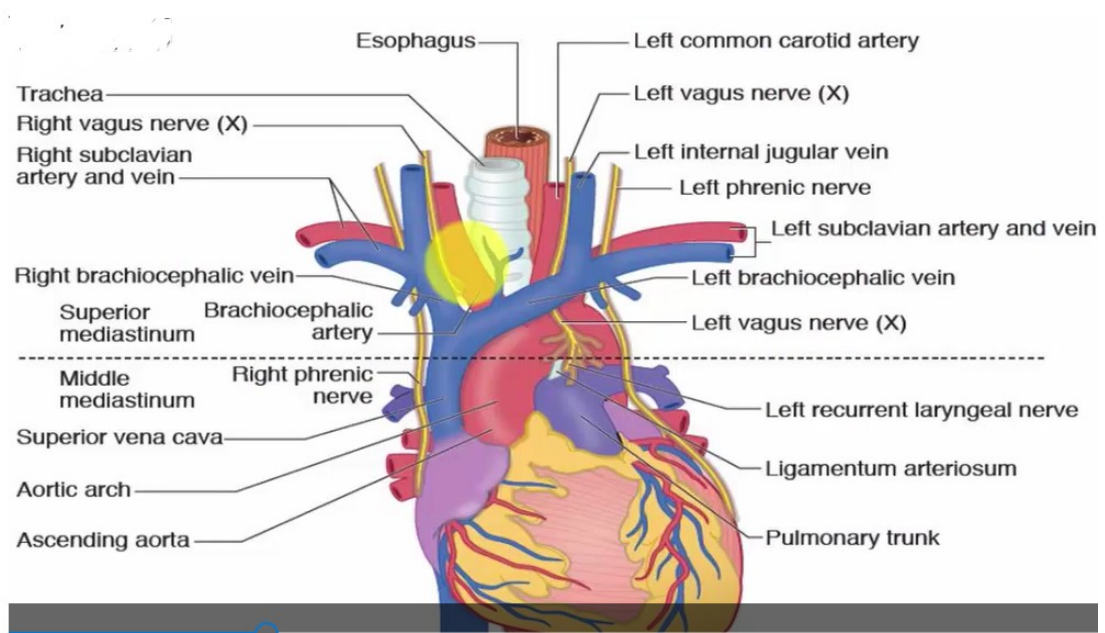
Right phrenic nerve

- Passes along the **right** side of the **right brachiocephalic vein**, **SVC**, and the **pericardium** over the right atrium.
- It also passes **anterior** to the **root** of the **right lung** and descends on the right side of the IVC to the diaphragm which it pierces near the **caval opening**



Left phrenic nerve

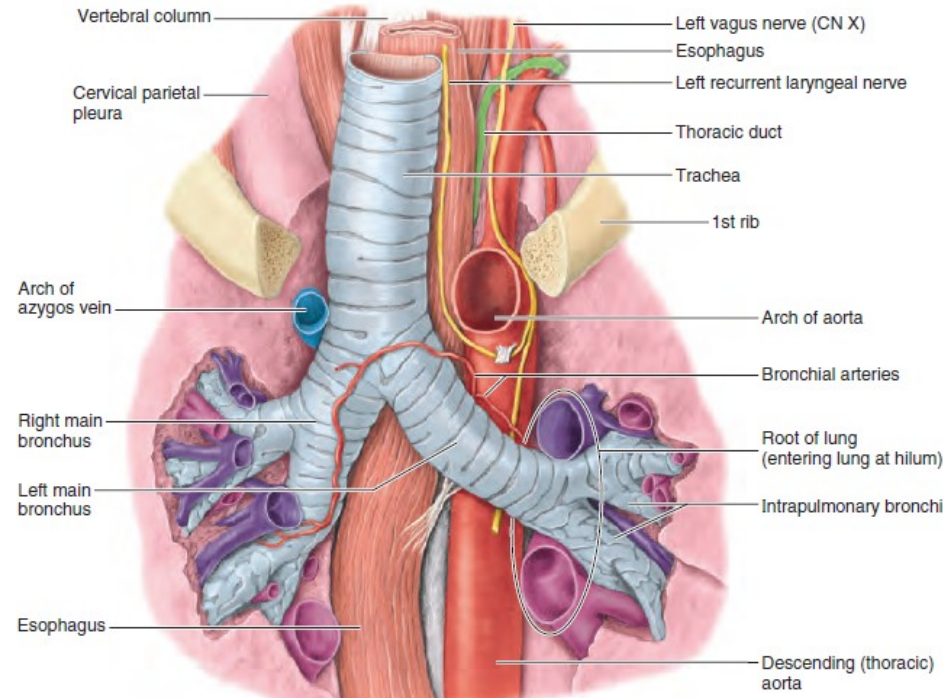
- It descends between the **left subclavian** and **left common carotid arteries**.
- It crosses the left surface of the arch of the aorta
- Then it descends anterior to the root of the left lung and runs along the fibrous pericardium and pierces the diaphragm.



Visceral plane

Trachea

- It descends **anterior** to the **esophagus**
- Enters the superior mediastinum, inclining a little to the **right** of the **median plane**.
- The posterior surface of the trachea is flat where it is applied to the esophagus
- The trachea ends at the level of the **sternal angle** by dividing into the **right** and **left main bronchi**.
- The trachea terminates superior to the level of the heart and is **not a component of the posterior mediastinum**



Esophagus

- The **esophagus** is a fibromuscular tube that extends from the pharynx to the stomach.
- It enters the superior mediastinum between the trachea and vertebral column, where it lies anterior to the bodies of the T1–T4 vertebrae.
- Initially, it **inclines** to the **left** but is pushed back to the median plane by the **arch of the aorta**.
- the *thoracic duct* usually lies on the left side of the esophagus, deep (medial) to the arch of the aorta.

